CMPSC 345 Lecture 5: Process Models Cheatsheet

1. SDLC Overview

SDLC: Structured process to design, develop, & test software.
Goal: Ensure systematic, efficient delivery & maintenance.
Stages: Planning → Defining → Design → Dev → Test → Deploy.

2. Code-and-Fix ("The Simplest")

Cycle: Problem → Code → Compile → Test → Release.
Pros: Good for tiny, one-off tasks (scripts).
Cons: No structure, messy code, bugs increase, hard to maintain.

[QUIZ SCENARIO]: Student writes code, runs it, fixes errors, repeats without planning. Ans: Code-and-Fix.

3. Waterfall Model (Royce 1970)

Key Feature: Sequential/Linear. Each stage must complete before next starts.
Flow: Plan → Req → Design → Impl → Test → Maint.
Pros: Clear structure, docs at every stage, easy to track.
Cons: Hard to change (no going back), late bug discovery.

[QUIZ NOTE]: Disadvantages = "Very little overlapping, single iteration, sequential order".

4. Incremental Model

Concept: Break project into components. Release in versions.
- **Release 1:** Core functionality.
- **Release 2+:** Add features/fixes.

Pros: Early delivery, visible progress, lower risk than Waterfall.
Cons: Needs good upfront arch, integration complexity.

[QUIZ SCENARIO]: Startup wants to release basic product quickly then add features. Ans: Incremental.

5. Spiral Model (Boehm 1988)

Key Driver: RISK ANALYSIS (Not time or features).
4 Quadrants (Iterative):
1. Determine Objectives/Constraints.
2. **Evaluate Alternatives & Resolve Risks.**
3. Develop & Test (Prototyping).
4. Plan Next Phase.

Pros: Risk reduction, flexibility.

[QUIZ SCENARIO]: Medical device/High safety risk, expensive failures. Ans: Spiral.

6. Rational Unified Process (RUP)

Framework: Iterative, Use-case driven, Architecture-centric.
4 Phases (Time-based):
1. **Inception:** Scope ('Should we?').
2. **Elaboration:** Arch & Risks ('Can we?'). *Critical Reqs*.
3. **Construction:** Building ('Building correctly?').
4. **Transition:** Deployment ('Ready for users?').

[QUIZ NOTE]: Objective to establish 'major/critical requirements'? Ans: Elaboration.

Summary Table

Model	Best For	Key Feature
Code-and-Fix	Homework, Scripts	No process
Waterfall	Stable Reqs	Sequential
Incremental	Early Value	Multi-Release
Spiral	High Risk/Safety	Risk-Driven
RUP	Large Enterprise	Use-Case/Arch